



Wibral-line PID for physical intelligence

Results, limits and the next sensor experiment

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The immediate application is **offline sensor analysis for embodied prediction**: two cameras, with optional audio, used to predict a separately measured future physical outcome. This is an attainable first step toward robust world models and robot policies. It is not yet a demonstrated training or control improvement.

The selected functional is **Makkeh–Gutknecht–Wibral categorical shared exclusions (MGW)**, in nats, with signed atoms. The defining paper is distinct from continuous shared exclusions and from other PID definitions. Fitted bins define a categorical estimand; raw sensor information is a different object.

The practical finding

An extra sensor's value is not its synergy alone. For two-source atoms R, U_1, U_2, S , the added Shannon information is $U_2 + S$. If source 1 is available with probability a and source 2 with probability b , with availability indicators M_1, M_2 such that $M_1 \perp M_2$ and $(M_1, M_2) \perp (X_1, X_2, Y)$, the added Bayes log-loss value is

$$\Delta = b[(1 - a)R + U_2 + aS].$$

Both predictors must see the same mask, and the complete-data policy and target law must remain fixed. Redundancy can provide backup value; unique information and synergy combine when sensors are present. The full proof includes correlated failures and data-dependent missingness in the detailed report.

For independent fair bits A, B, U , set the same target $Y = (A, B)$ and baseline $X_1 = A$. Candidate noise U and useful complement B both have MGW synergy $\ln(4/3)$. Their added information is respectively zero and $\ln 2$. A duplicate A has zero synergy but supplies backup value when the first copy is unavailable.



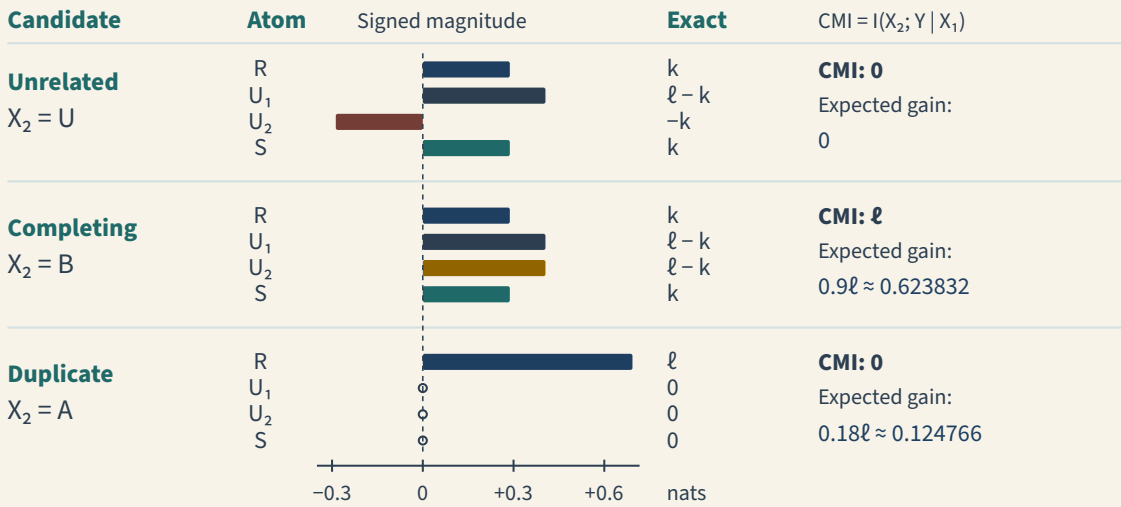


Same synergy, different added information

Exact analytic example · signed net MGW atoms in nats

A, B, U independent fair bits; target $Y = (A, B)$; baseline $X_1 = A$.

$\ell = \ln 2 \approx 0.693147$; $k = \ln(4/3) \approx 0.287682$



U and B have the same synergy k , but their added information is 0 versus $\ln 2$.

Gain: $a = 0.8$, $b = 0.9$; observed availability masks independent of data and each other.

Bayes log-loss information gain; learned-policy benefit is unproved. Zero atoms use dots.

Figure 1: One target law, three candidate sensors. Signed unique information distinguishes noise from a useful complement despite equal synergy; duplication can have backup value.

There is also a useful limit: the expected Bayes log-loss reduction relative to the no-source predictor, under a fixed law and exogenous availability, depends only on subset mutual information. For two to four sources, that is 3, 7 or 15 values. Full PID's 4, 18 or 166 atoms provide a finer allocation, but no extra decision information for that exact objective. PID must earn additional cost through a useful diagnostic question or a separately justified objective.

What exists and what it supports

Work	Useful role	Remaining boundary
Rust categorical MGW for 2–4 sources	Offline signed atom calculation with budgets and cancellation	No completed sensor benchmark follows from library availability
Equal-synergy and target-copy formal work	Reject an invalid synergy-only selection rule	Exact theorem premises and replay status remain in the theorem maps
Fixed-alphabet continuity and dependence-aware law bounds	Conditional tools for estimator sensitivity	A valid sampling/dependence model is still required
Finite-prefix mean, bias and gradient work	Mathematical tools for a future bounded encoder experiment	Not an implemented robot-training method
New availability exposition	Complete classical derivation, controls and experiment design	Handwritten application synthesis; no new Lean probability theorem or scientific-priority claim

The mathematical results guide points to standalone proofs and exact formal status. A verified mathematical intermediate step does not establish estimator calibration or physical value.





Rust implementation and computational cost

Rust already supplies categorical MGW PID for two to four sources through the [averaged, budgeted API](#). The proposed sensor adapter and learning loop are not implemented by this study. With n source groups, N rows, K occupied complete states, d categorical columns and $m \in \{4, 18, 166\}$ atoms, the per-node event work is quadratic in K . Averaged output saves retained pointwise results, while row and PMF storage remain. Feature extraction, quantization, up to 2^n masked predictions per landmark, model training and repeated resampling have separate costs. The first application should run offline. No end-to-end latency or learning benefit has been measured here; see the [full implementation and cost discussion](#).

The ecosystem connection

CREBAIN produces observations; NCP transports typed sensor records; Prisoma records and inspects experiments; pid-rs computes the declared information quantities; Galadriel can retain diagnostic evidence. Manwe supplies a separate tracking/measurement setting. Existing consumers and proposed connections are distinguished in the detailed report. No protected checkout was changed.

The current two-camera-plus-audio example records five images and 800 pressure samples over six 120-Hz body ticks. That proves a capture composition, not independent statistical replication. The missing step is a frozen feature table joined to future reference labels. Use one declared episode/landmark unit, preserve timing and missingness, fit transforms only on training episodes, and evaluate untouched episodes.

How practical value will be tested

Start offline with all eight sensor subsets. Measure predictive loss, availability, runtime and memory. Keep fixed-model outages separate from retrained subset models. Compare MI/CMI, subset task loss and established masking methods. [Masked sensor training](#) and [MAD multi-view reinforcement learning](#) are relevant baselines; no superiority to them is established here.

The detailed report derives a conservative paired-episode loss bound and states its independence and bounded-probability assumptions. It also explains why natural occlusion needs conditional laws, why more sensors can defeat finite-data estimation, and why an action-conditioned latent must not be evaluated against its own injected action.

Closed-loop control, sensor placement and re-identification follow only after their target, policy, association and cost contracts are defined. The first acceptance criterion is a useful gain on a realistic task against matched baselines, with negative results retained. The mathematical goal is to explain and enable that gain, not to substitute a new score for evidence.

The detailed report explains how formal proof checking, bounded Rust tests, evidence-record audits and PDF review address different errors. These processes support the mathematical and engineering work; useful sensor performance still requires the proposed held-out episode evaluation.

How to cite: Sepehr Mahmoudian, *Wibral-line PID for physical intelligence*, pid-rs technical report, 2026. Include the exact repository commit or release. Use the [citation metadata](#) for the software and cite the defining method papers separately.

